# A Four-Island Multiarticle

## TISO, Certified GBG Self-Embeddings, Undecidable Attainment, and J4

From resource allocation to verifier-safe search, with a carefully bounded gravitational–quantum application

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Author-directed research synthesis; AI-assisted mathematical drafting and checking

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**See the AI disclaimer that follows this essay.** The complete AI-assistance, authorship, verification, and reliability declaration is the final page. The new mathematical statements below have paper-level arguments, not independently machine-verified certificates. The J4 physical problem is not claimed solved.

**Abstract**

We integrate exactly four source islands: Tenneson's settlement-tensor optimization (TISO), the generalized-board-game (GBG) major-theorems exposition, the finite-checkable but undecidable attained-labels manuscript, and the J4 gravitational–quantum progress report. The integration is not a union of abstracts. We construct explicit bridges at the level of typed states, certificates, costs, predictive models, and frozen physical premises. Three derived results are central. First, a nonnegative-cost address extension of a certified GBG preserves the infimum charged cost of an accepting proof, so the structural self-embedding alone gives no intrinsic shortest-proof speedup. Second, if the settlement tensor and feasible allocation data are pulled back along the proof projection, the TISO objective and its set of optimizers are exactly preserved. Third, the source's five-party probability example gives a sharp 1/4 obstruction to calibrating a particular event probability from unweighted outcome support alone. We combine these with verifier-gated adaptive conservativity, finite-budget reachability, and the limited classical-channel J4 no-go result. We state a low-fan-in extraction ledger (supports of size +1 through +6), identify the distinction between derivational synthesis and genuinely new closure-relative Multiarticle surplus, and leave realistic proof-search acceleration and the full J4 dynamical-gravity target open.

**Keywords.** Multiarticle Theory; generalized board games; TISO; certificate transport; proof cost; computational budgeting; MUSIC; SYMPHONY; undecidable labels; AMLD+0; J4; Einstein–Peres; DNI.

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# 1 Four frozen source islands and provenance

The islands are denoted $A_1, \ldots, A_4$ throughout. Section/theorem references identify the source version; all novel cross-source assertions are labelled *derived here* or *open*. The names of the islands are part of the frozen input, not proof that their union is logically stronger than each component.

| Island | Source | Mathematical role here |
|---|---|---|
| $A_1$ | *Discrete Settlement Tensors and Constrained Optimization in DATA 3.0 / AMLD—TISO* (16 Oct. 2026), especially §§13–18 [1] | State-indexed tensor, auditable attribution, concave static optimization, $2\varepsilon$ proxy guarantee and charged costs. |
| $A_2$ | *Major Theorems of Generalized Board Games: From a Certified Self-Embedding to Reachability, Board Equivalence, and Higher-Order Reports* (4 Nov. 2026), especially Theorem 2.1 and Corollary 2.2 [2] | Effective labelled move systems, verifier observations, address self-embedding, and projection-preserved traces. |
| $A_3$ | *Undecidable Attained Labels with Finite Checkable Witnesses*, audited ed. 2.0 (3 Nov. 2026), especially Theorems 4.1, 6.3, 7.2 [3] | Equal support/different probability, finite witnesses versus undecidable attainment, and verifier-safe adaptive composition. |
| $A_4$ | *Progress Toward the J4 Gravitational–Quantum Interface*, audited v2 (4 Nov. 2026), especially §§5,10–14 [4] | A frozen physical research target, restricted same-generator/no-go result, observability barriers, and explicitly synthetic budget-pivot tests. |

**The principal type distinction.** $A_1$ is a mathematical *navigation* model, $A_2$ an *operational proof-search* model, $A_3$ a *verification and computability* boundary, and $A_4$ a collection of *physical-model obligations* with a research procedure. An optimized search policy is not itself a proof of a physical statement.

**Terminology.** We preserve the author's six AMLD+0 report outcomes $P, R, D, I, C, U$ when referring to AMLD+0. TISO's repaired source sometimes uses $K$ for a contradiction-investigation role and treats $D$ as an investigative contribution rather than a terminal output. This essay does not perform a role or outcome migration: a TISO channel name and an AMLD+0 certified report are distinct typed objects.

# 2 A typed Multiarticle operation, not textual concatenation

**Definition 2.1** (Source translations and consequence bridges)**.** For source island $A_i$, let $\Gamma_i$ be its explicitly licensed mathematical claims with their hypotheses and proof statuses. A candidate Multiarticle chooses a fixed comparison language and consequence relation $\vdash_F$, together with partial meaning-preserving translations $\tau_i$ and declared bridging assumptions $\mathcal{B}$. The combined closure is

$$\mathcal{C}_M = \mathrm{Cn}_F\left(\bigcup_{i=1}^{4} \tau_i(\Gamma_i) \cup \mathcal{B}\right). \tag{1}$$

A candidate $\varphi$ is *closure-relative surplus* over all four source islands only if $\varphi \in \mathcal{C}_M$ and $\varphi \notin \mathrm{Cn}_F(\tau_i\Gamma_i)$ for every $i$. A genuinely source-essential claim additionally requires adequate source-removal tests.

The non-derivability side is not established by failure of a proof search. A bridge that simply assumes the desired target does not establish genuine integration. Unless complete source closures or suitable countermodels have been audited, this essay uses *derived cross-source consequence*, not *certified transcendent theorem*, for its new results [3, 4].

**Definition 2.2** (Proof-support fan-in)**.** A labelled $+d$ extraction is a displayed conclusion with an explicit proof/dependency sketch citing at most $d$ distinct source facts or prior lemmas, with declared types and premises. This is an *upper bound* on local support size, not a certificate of minimal support, novelty, or independence. Because there are only four islands, $d = 5, 6$ refers to facts/lemmas, not five or six distinct articles.

# 3 The exact GBG bridge: proof projection and no free shortest-proof speedup

Fix a deterministic, verifier-labelled proof GBG $H = (Q, q_0, L, (r_\mu)_{\mu\in L}, V_H, c_H)$, with finite move alphabet, partial effective transitions and nonnegative edge charges. Let $\mathrm{Acc}_H(q)$ mean that $V_H$ accepts a proof certificate for the same frozen decoded target. Assume the acceptance observation is determined by the state, and that all proof-relevant verification costs are included in $c_H$ or in an identical terminal verification charge.

Following the constructive address extension in $A_2$, let $W = \{0,1\}^*$ and $G = W \times Q$, with address actions $a_b(w, q) = (wb, q)$, lifted proof actions $\widetilde{r}_\mu(w, q) = (w, r_\mu(q))$, projection $\pi(w, q) = q$, and proper embedding $e(w, q) = (0w, q)$. Accept at $(w, q)$ exactly when $\mathrm{Acc}_H(q)$; charge lifted edges their original costs and every address edge a declared nonnegative cost.

**Lemma 3.1** (Projection and lifting; derived here from $A_2$)**.** *Deleting address moves from any legal finite $G$ path from $(\epsilon, q_0)$ produces a legal $H$ path from $q_0$ with the same terminal certificate status and no larger charged cost. Every finite $H$ path from $q_0$ has a legal, equally charged lift to $G$ containing no address move.*

*Proof.* Every address action leaves the $Q$ coordinate fixed; each lifted proof action changes $Q$ exactly as its $H$ antecedent. Deleting address actions therefore gives a legal proof-move history. Acceptance depends only on $Q$ and nonnegative address costs can only increase total charge. For the converse, keep the address word $\epsilon$ and lift each $H$ action. All lifted edges retain their original charges and acceptance view. □

**Theorem 3.2** (No intrinsic shortest-proof acceleration from address extension; derived here)**.** *Let $d_H$ and $d_G$ be the infima of total charged costs among accepting finite paths from $q_0$ and $(\epsilon, q_0)$, respectively, with* $\inf \emptyset = +\infty$*. Under the preceding contract,*

$$d_G = d_H. \tag{2}$$

*Starting instead from the transported root $e(\epsilon, q_0) = (0, q_0)$ also gives the same infimum* after *that root is reached; reaching it from the original root may incur a setup charge.*

*Proof.* Lemma 3.1 maps every accepting $G$ path to an accepting $H$ path of no greater cost, so $d_H \le d_G$. Its lift maps every accepting $H$ path to a $G$ path with equal cost, so $d_G \le d_H$. The proof remains valid for empty acceptance sets. Transported-root comparisons use the same projection argument; any compulsory address setup from $(\epsilon, q_0)$ must be counted separately. □

*Remark* 3.3. This is a statement about the *intrinsic best certificate path*. A resource-limited or poorly informed scheduler may search paths in a different order, and a representation change may improve its empirical time to a proof. Such an improvement requires a separate algorithm, comparator, and full accounting; it does not follow from self-embedding alone. The underlying GBG theorem is constructive, not universal for arbitrary prior games [2].

## 4 Pulling a settlement tensor across a GBG embedding

For a finite set of channels $J$, TISO first converts state-dependent, deduplicated and attributed tensor events into weights $W_j(x) \geq 0$. It then considers, for frozen $k_j > 0$, feasible set $\mathcal{U}$ and complete charges,

$$H_x(u) = \sum_{j \in J} W_j(x)(1 - e^{-k_j u_j}), \qquad F_x(u) = 1 - e^{-H_x(u)}. \tag{3}$$

The second expression is an optimizable surrogate; it is a settlement probability only if the source's additional, unvalidated Poisson-opportunity model holds [1, §14].

**Theorem 4.1** (TISO equivariance under projection-invariant features; derived here)**.** *Assume the tensor feature and event-identity convention on the address extension is a pullback, namely $S_G(w, q) = S_H(q)$, where copies of one accepted event retain a single identifier. Suppose calibration, $J$, $(k_j)$, charged costs and $\mathcal{U}$ are fixed identically at projected states. Then for all $u \in \mathcal{U}$,*

$$F^G_{(w,q)}(u) = F^H_q(u), \qquad \operatorname{argmax}_{u \in \mathcal{U}} F^G_{(w,q)}(u) = \operatorname{argmax}_{u \in \mathcal{U}} F^H_q(u). \tag{4}$$

*If $\mathcal{U}$ is nonempty and compact, both optimizer sets are nonempty. If the common objective is strictly concave on $\mathcal{U}$, they consist of the same unique optimizer.*

*Proof.* The frozen projection-invariance assumptions make every term $W_j, k_j, u_j$ in (3) agree, so the intensities and objectives coincide pointwise. The feasible sets are identical, hence so are the maximizer sets. The existence and uniqueness assertions follow from ordinary continuity and strict concavity, as in TISO Theorem 15.1 [1]. □

*Remark* 4.2 (Necessity of the extra hypothesis). Certified GBG self-embedding alone does *not* imply (4). For instance, a tensor estimate depending spuriously on the address length, $W(w, q) = 1 + |w|$, changes the TISO objective under $e$ even though the verifier sees the same proof component. This is a counterexample to an *unqualified* tensor-invariance assertion, not an objection to Theorem 4.1.

## 5 Why support-only MUSIC cannot calibrate a TISO probability

The third island constructs two five-qubit preparations $\rho_+$ and $\rho_-$ with identical proper-subsystem reduced states and full support for each of the 32 local $X$-measurement outcomes, but different joint laws. If $P(x) = \prod_{i=1}^5 x_i \in \{+1, -1\}$, their exact probabilities are

$$p_\sigma(x) = \frac{1}{32}\left(1 + \frac{\sigma P(x)}{2}\right), \quad \sigma \in \{+1, -1\}. \tag{5}$$

Thus $\Pr_+(P = +1) = 3/4$ and $\Pr_-(P = +1) = 1/4$; their unweighted supports agree [3, Thm. 4.1].

**Theorem 5.1** (Support-only calibration obstruction; derived here from $A_1, A_3$)**.** *Suppose a predictive rule for the event $E = \{P = +1\}$ receives only the unweighted set of permitted outcomes (or its full powerset), with no preparation tag or probability information. It must return the same scalar $z$ for the two preparations in* (5)*, and*

$$\max\left\{\left|z - \frac{3}{4}\right|, \left|z - \frac{1}{4}\right|\right\} \geq \frac{1}{4}. \tag{6}$$

*Consequently support alone cannot establish a uniform probability-calibration error $\varepsilon < 1/4$ on this two-instance family for that event. A TISO success-probability interpretation or its $2\varepsilon$ expected-utility bound needs genuinely probability-bearing evidence and the source's own additional assumptions.*

*Proof.* Full support is identical under $\rho_+$ and $\rho_-$, so a function only of that support has the same output $z$. By the triangle inequality, $\frac{1}{2} = |\frac{3}{4} - \frac{1}{4}| \leq |\frac{3}{4} - z| + |z - \frac{1}{4}| \leq 2\max\{|\frac{3}{4} - z|, |z - \frac{1}{4}|\}$. Equality is attained at $z = 1/2$. The calibration consequence is about the specified parity event and identical input encoding, not about all possible TISO predictors. □

*Remark* 5.2 (Physical scope)*.* This is a finite five-party quantum *information obstruction*, not a J4 two-qubit gravitational prediction. Its use for J4 is methodological: an experiment-selection or proof-search controller should not treat the support of possible observations as if it determined the law of outcomes.

## 6 Finite resource reachability versus unbounded undecidability

**Theorem 6.1** (A decidable bounded horizon does not decide general attainment; derived here)**.** *In an effectively presented finite-labelled GBG with decidable state transitions, acceptance and cost, the existence of an accepting run of length at most $n$ is decidable by finite enumeration. If every transition costs at least a known $\delta > 0$, then existence of an accepting run of total cost at most $B$ is likewise decidable by enumeration through length $\lfloor B/\delta \rfloor$. Nevertheless, across effectively presented GBGs, existence of* any *accepting finite run is undecidable, even though each presented successful run can be checked in finite time.*

*Proof.* A finite move alphabet gives finitely many label words through any fixed finite depth; each is effectively simulated and checked. Positive minimal cost bounds the length of every budget-feasible run. For unbounded attainment, encode a program $e$ as a deterministic proof-search system whose sole move simulates one machine step and whose accepting predicate holds exactly after the program halts. A finite accepting trace is mechanically checkable, but deciding whether one exists for every $e$ would decide the halting problem. This is the finite-witness obstruction articulated in $A_3$ [3, Thm. 6.3]. The address extension and TISO resource scores do not change that reduction. □

*Remark* 6.2*.* The conclusion is about a *family of effective formal searches*; it does not assert that the particular, presently formulated J4 question has been proved undecidable. A finite budget exhausted without a certificate is not a proof of independence or a certified closed basin.

## 7 Verifier-safe adaptive TISO allocation

Let $\Gamma$ and formal calculus $F$ be fixed; let $B_t$ contain only checker-accepted, scope-valid mathematical statements at stage $t$, and let $C_t$ be an allocation/composer policy. The third island allows $C_{t+1} = \Phi(C_t, S_t, B_t)$ provided the independent checker, proof-admission criteria and frozen $\Gamma$ are not modified [3, Thm. 7.2].

**Theorem 7.1** (Certified adaptive invariance; cross-source specialization)**.** *Suppose the controller chooses each next allocation by optimizing a frozen, feasible TISO surrogate on a GBG state; after the resulting search, only statements carrying sound checker-accepted derivations from* $\Gamma \cup B_t$ *enter* $B_{t+1}$*. If* $B_0 \subseteq \mathrm{Cn}_F(\Gamma)$*, then for every finite* $t$*,*

$$B_t \subseteq \mathrm{Cn}_F(\Gamma), \qquad \mathrm{Cn}_F(\Gamma \cup B_t) = \mathrm{Cn}_F(\Gamma). \tag{7}$$

*The conclusion also holds on the projected address-extension GBG, provided the verifier and dependency records are preserved.*

*Proof.* Induction on accepted deposits: a newly validated proof from $\Gamma \cup B_t$ can substitute the previously accepted derivations of $B_t$ from $\Gamma$. It is consequently derivable from $\Gamma$ alone. Closure monotonicity and idempotence imply (7). The TISO choice changes only the order or resource allocation of attempts, not the checking rule. The address projection preserves the accepted evidence and scopes by hypothesis. This is a typed specialization of the source's verifier-gated conservativity theorem, not a new general soundness principle. □

**What this theorem does not say.** It does not show that the adaptive controller has a calibrated success probability, outperforms all baselines, always terminates, or solves an undecidable question. TISO's source explicitly excludes such inferences. Cross-agent logical AND effects can also invalidate a separable concavity model [1, §17].

# 8 The J4 import: a frozen physical no-go is not escaped by rescheduling

The fourth island separates a genuine gravitational source requirement from a narrower, nonrelativistic reciprocal measurement-and-classical-feedback model. For this restricted model class $\mathcal{M}_{\mathrm{cl}}$, a source-derived algebraic identity under declared balance assumptions yields

$$ab - g^2 - k^2 = (xy - \nu\xi + \alpha\beta)^2 + (x\beta - y\alpha)^2 \geq 0, \tag{8}$$

with $g = x\nu + y\xi$, $a = x^2 + \alpha^2 + \xi^2$, $b = y^2 + \beta^2 + \nu^2$ and $k = \alpha\nu + \beta\xi$ in the stated convention. Here the *physical noise-covariance parameter* $k$ is not TISO's search-efficiency parameter $k_j$. The report emphasizes that this effective classical-channel construction is not a generally covariant, backreacting source theory [4, §5].

**Proposition 8.1** (No-go preservation by sound scheduling; derived cross-source corollary)**.** *Let* $\Gamma_{\mathrm{cl}}$ *consistently formalize the restricted model assumptions and the theorem* (8)*. Suppose a GBG/TISO/AMLD+0 search keeps these premises and target interpretations frozen and accepts only sound proofs. Then no accepted, same-model certificate may establish the strict contrary inequality* $g^2 + k^2 > ab$ *for a member of* $\mathcal{M}_{\mathrm{cl}}$*. A successful proof about a model* outside $\mathcal{M}_{\mathrm{cl}}$ *requires new separately declared hypotheses and checks; it cannot be inferred merely from a changed search allocation.*

*Proof.* The inequality (8) is a nonnegative sum of squares within the restricted class. Its strict contrary would contradict it under identical hypotheses and parameter semantics. In a consistent theory, a sound verifier cannot accept both. Theorem 7.1 ensures that modifying the search scheduler admits no unlicensed new premise. It does not bar a correctly declared theory migration to a different physical class. □

**J4 remains open.** The decisive target requires a single physically admissible gravity–matter dynamics $T$ and a controlled reduction yielding the joint interaction/noise/observation data $(g, a, b, k)$ from that *same* dynamics, with causal, positivity, field and backreaction obligations. The existence or impossibility of a general covariant solution is not established by this four-island synthesis [4].

## 9 Low-fan-in extraction ledger: +1 through +6

The following table is a transparent dependency map, *not* a claimed minimal-fan-in computation or complete enumeration of logical closure. A "+" count is a convenient support-size cap; from four islands one can cite more than four distinct proven source facts.

| Label | Derived or source-grounded result | Explicit support and status |
|---|---|---|
| +1 | Conditional static TISO concavity/global optimum; or finite-witness undecidable attainment. | Each originates within one source island: $A_1$ Thm. 15.1 or $A_3$ Thm. 6.3. *Source-derived.* |
| +2 | Projection-preserving GBG addresses give equal TISO optimizers if the features pull back (Thm. 4.1); or identical support obstructs a single probability predictor (Thm. 5.1). | $A_2$ embedding + $A_1$ TISO; alternatively $A_3$ parity probabilities + $A_1$ calibration requirement. *Derived here.* |
| +3 | Tensor-guided allocation can be adapted while retaining verifier-safe BANK conservativity (Thm. 7.1). | $A_1$ allocation, $A_2$ verifier-labelled transitions, $A_3$ frozen-contract induction. *Cross-source specialization.* |
| +4 | Restricted J4 contradictions cannot be made certifiable by TISO scheduling on a certified GBG (Prop. 8.1). | $A_4$ sum-of-squares theorem, $A_3$ BANK discipline, $A_2$ proof transport, $A_1$ navigation/feasibility only. *Derived scoped corollary*; some supports are contextual rather than logically essential. |
| +5 | Combining equal intrinsic proof cost, optimizer equivariance and certified adaptation gives the joint assertion: a copied search has no intrinsic proof-distance gain, a common TISO optimum, and unchanged accepted deductive closure. | Lemma 3.1, Thms. 3.2, 4.1, 7.1, plus fixed common-budget semantics. *Conjunction of proved claims*, not a new irreducible five-premise theorem. |
| +6 | Add to the +5 package that finite-budget acceptance is decidable under effective positive-cost hypotheses, but unbounded acceptance need not be decidable. | The five facts in the preceding row plus Thm. 6.1. *Conjunction of proved claims*, not a new irreducible six-premise theorem. |

**Why the last two labels matter despite the limitation.** They display how increasingly rich *verified theorem packages* can be put into one common proof-support graph without claiming new logical dependence from mere aggregation. Genuine +5 or +6 *irreducible* surplus requires a specific target and countermodels or other non-derivability evidence for lower-support closures. Those certificates are absent here.

## 10 What the zip changes for J4—and what it cannot change

The four-island chain proposes an operational, auditable search architecture:

1. **Freeze the physical question.** Define the exact J4 model class, observable law, four-parameter reduction if applicable, and all gravity–matter consistency checks.
2. **Encode a typed proof board.** Use GBG legal actions for model derivation, counterexample generation, source-premise imports and independent certificate checking; never identify a candidate with an accepted proof.
3. **Compute transparent resource estimates.** Derive TISO weights from attributed, deduplicated evidence; keep calibration data separate from any hidden answers and charge computation, communication, verification, switching and representation costs.
4. **Allow certified-policy adaptation.** Player Zero may change feasible allocations and initiate source-justified bridge attempts while the checker keeps the underlying theory and proof admission frozen.
5. **Require probability-bearing experimental comparisons.** An observed set of possible scores or reports does not substitute for a law, preparation, or measurement context; test identifiability and error bounds explicitly.
6. **Report only what is witnessed.** A finite verified proof or counterexample may settle a frozen claim; exhaustion of an unbounded search budget is not an independence certificate, and a toy-grid scheduler win is not a physical J4 discovery.

The J4 report's 36-case finite-graph comparison (frozen TISO 20, charged certified escape 27, provisional pivot 29 verified goals under its stated 12-unit artificial budget) is useful as a *test fixture*, not evidence of live ATP superiority. Its own report supplies a counterexample to universal dominance of basin escape [4, §12].

## 11 Open, nontrivial research targets

**O1: An informative self-embedding.**
Exhibit a certified GBG self-embedding with a fully charged, equal-information policy improvement against a precisely defined strong baseline, or prove a corresponding impossibility under explicit invariance restrictions. The address-only result above supplies a negative control.

**O2: Support-sufficient calibration.**
Characterize report features and probability-bearing scores sufficient for a uniform TISO calibration guarantee. The support-only 1/4 obstruction shows that one natural impoverished encoding fails.

**O3: Beyond separable TISO.**
Define and analyze cross-agent AND/bridge complementarities with exact event credit and verify that the optimization algorithm is appropriate to the resulting potentially nonconcave objective.

**O4: Irreducible Multiarticle surplus.**
Select a fixed mathematical target and furnish a joint proof, source-specific nonderivability/countermodels and non-vacuous bridge assumptions. A theoremgraph union, even with the support ledger above, is not that certificate.

**O5: The physical J4 bridge.**
Supply one covariant and physically admissible joint gravity–matter source and a controlled observable reduction, or a precisely scoped impossibility theorem for a declared source class. Proof-search scheduling cannot stand in for this physical ingredient.

## 12 Conclusion and theorem-status declaration

This Multiarticle yields a concrete, checkable *research synthesis*: a no-free-shortest-proof-cost theorem for address-only GBG enlargement, a conditional TISO optimizer-equivariance theorem, a support-only 1/4 calibration barrier, a finite-versus-unbounded attainment distinction, and a verifier-safe J4 no-go inheritance corollary. The strongest conceptual link is the division of labor between *what is provable*, *where search is allowed to go*, *how finite resources are allocated*, and *what a physical model actually predicts*. No authentic formal proof-assistant receipt, complete-source non-derivability certificate, calibrated TISO probability law, live federation benchmark or solution of full gravitational J4 is asserted. All proof arguments here require independent mathematical review before publication.

## References

[1] Brian Tenneson, *Discrete Settlement Tensors and Constrained Optimization in DATA 3.0 / AMLD—TISO: Settlement Density, One-Hot Terminal States, Resource Allocation, and the Tensor-Induced Settlement Objective*, expanded and ATLAS-repaired research manuscript, 16 Oct. 2026, 15 pp. `https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/discrete_settlement_tensors_tiso_2026/`.

[2] Brian Tenneson, *Major Theorems of Generalized Board Games: From a Certified Self-Embedding to Reachability, Board Equivalence, and Higher-Order Reports*, research exposition, 4 Nov. 2026, 13 pp., user-supplied PDF and LaTeX source. See also the earlier GBG/GBG+ foundations and CAMI equivalence manuscripts cited therein.

[3] Brian Tenneson, *Undecidable Attained Labels with Finite Checkable Witnesses: Five-Party Quantum Separation, Deterministic Realization Limits, and Verifier-Gated Adaptive Composition*, Atlas-audited research exposition 2.0, 3 Nov. 2026, 14 pp., user-supplied PDF.

[4] Brian Tenneson, *Progress Toward the J4 Gravitational–Quantum Interface: Multiarticle Theoremgraphs, Deterministic–Nondeterministic Interfaces, AMLD+0, Basin Escape, and Resource-Constrained Discovery*, Atlas-audited v2, 4 Nov. 2026, 19 pp., user-supplied PDF. Includes attribution of its restricted same-generator classical-channel result to a separate cited manuscript.

[5] Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. Standard mathematical background for the convex-programming results cited by TISO.

## Final-page AI-assistance and verification disclaimer

This research exposition was prepared with AI assistance in response to Brian Tenneson's request to combine four named research articles using his Multiarticle methodology. The four source islands, research direction, and intended J4 application are user-selected. ChatGPT (GPT-6) drafted the present integration, specified bridging hypotheses, supplied or reconstructed the displayed paper-level mathematical arguments, organized the support-size ledger, and produced the LaTeX source and a PDF rendered from that same source. Results attributed to the source manuscripts remain their respective authors' results; the new cross-source statements and proof sketches are internal AI-assisted derivations, not externally certified mathematical discoveries or claims of world-first originality. Standard background claims (including the halting problem, convex optimization, and consistency reasoning) are not presented as original to this work.

No independent proof-assistant verification, authenticated AMLD+0 BANK receipt, deployed five-agent ATP federation, physical experiment, calibration study, verified Poisson arrival model, general speedup, full-source closure nonderivability check, or gravitational–quantum unification is asserted. In particular, TISO's mathematical optimum is a frozen-model surrogate optimum; it is not necessarily the true success probability or the optimal dynamic search policy. The GBG no-speedup conclusion applies to the explicit address extension and its fully charged acceptance/cost projection, not arbitrary self-referential machines. The quantum probability obstruction uses a five-party toy preparation, not a J4 experiment. The physical J4 conclusion is restricted to the class and assumptions explicitly stated. The report codes $P, R, D, I, C, U$ must not be reassigned without authorization. A claimed "settlement" requires its declared verification contract, and lack of a found proof is not a refutation or independence certificate.

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